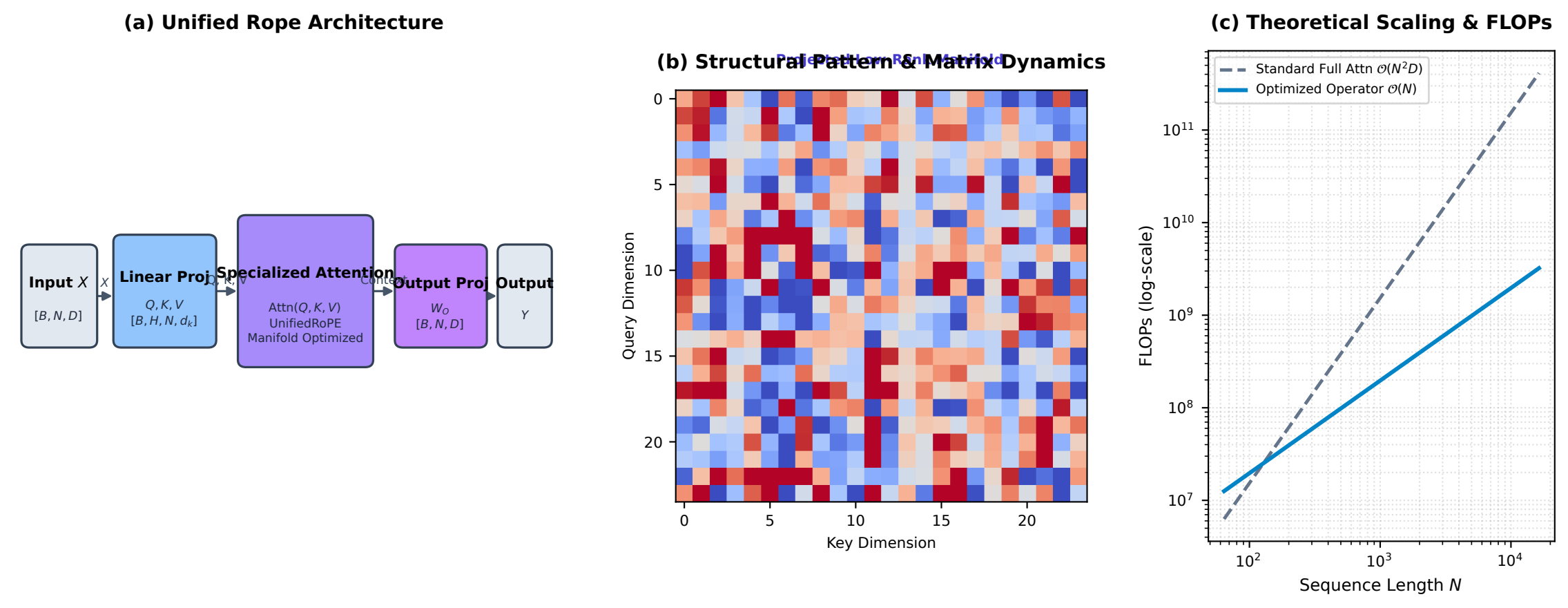
(a) Unified Rope Architecture
Input X
$[B, N, D]$
X
Linear Proj
Q, K, V
$[B, H, N, d_k]$
Q, K, V
Specialized Attention
Attn(Q, K, V)
UnifiedRoPE
Manifold Optimized
Context
Output Proj
W_O
$[B, N, D]$
Output
Y
(b) Structural Pattern & Matrix Dynamics
Query Dimension
Key Dimension
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Optimized Operator $\mathcal{O}(N)$
FLOPs (log-scale)
Sequence Length N
10^7
10^8
10^9
10^{10}
10^{11}
10^2
10^3
10^4